

Developing a Cross-National Framework for Comparing Predictors of Compliance with COVID-19 Protective Behaviours Before and After Mask Mandates

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Abstract

This work investigates how predictors of compliance with COVID-19 protective behaviours differ before and after the introduction of mask mandates, and whether these predictive patterns can later be compared across national settings. The analysis adopts a predictive rather than causal perspective: the aim is not to estimate the causal effect of mask mandates, but to identify which demographic, behavioural, health-related, attitudinal, and temporal variables are most informative for predicting compliance in different policy periods.

Australia is used as an initial case study to develop a reproducible data preparation and modelling pipeline before extending the analysis to additional countries. The analysis links repeated cross-sectional survey data from the Imperial College London/YouGov COVID-19 Behaviour Tracker with policy timing data from the Oxford COVID-19 Government Response Tracker. After cleaning, recoding, and linking the two sources by state and date, the Australian analytical dataset contains 35 460 observations from 2 September 2020 to 1 March 2022. Two binary outcomes are analysed: face mask wearing and broader protective behaviour. Four classification models are compared—logistic regression, classification tree, random forest, and XGBoost—using stratified train–test splits, cross-validation on the training data, and held-out test evaluation.

Initial results show that XGBoost achieves the strongest predictive performance across the four Australian modelling tasks, and that the relative importance of predictors differs between pre-mandate and post-mandate periods. These findings provide a basis for extending the same analytical framework to later cross-national comparison.

Keywords: COVID-19; protective behaviours; mask mandates; compliance; machine learning; XGBoost; Australia; cross-national comparison

1 Introduction

Protective behaviours such as face mask wearing, distancing, hand hygiene, and self-isolation were central to public health responses during the COVID-19 pandemic. These behaviours were important because they helped reduce transmission risk when pharmaceutical interventions were unavailable, limited, or unevenly distributed. However, adherence to protective behaviours was not shaped by policy rules alone. Previous studies have shown that compliance was also associated with demographic characteristics, perceived risk, psychological factors, social context, and broader behavioural patterns [1–5]. Understanding which factors predict compliance is important because these factors may help identify targets for public health communication and behavioural interventions during future infectious disease outbreaks [1, 6].

Mask mandates provide a useful setting for studying changes in the predictors of compliance. Mandates may alter not only the overall level of mask wearing, but also the relative importance of the factors that explain who complies and who does not. Before mandates, mask wearing may be more strongly associated with voluntary risk perception, trust, or individual attitudes. After mandates, compliance may instead be more strongly structured by policy context, social expectations, or willingness to follow public health advice. However, few studies have focused on whether there are common predictive patterns before and after the implementation of mask-wearing policies across different countries [6].

Building on Ryan et al. [6], which found that mask mandates altered major determinants of adherence to protective health behaviours in Australia, this project extends the analysis toward a broader cross-national comparison.

The broader aim of this project is to construct a reproducible pipeline that can be applied consistently across countries. Australia is used first because it provides a useful case for developing and testing the workflow: mask policies varied across states, the YouGov survey provides repeated cross-sectional behavioural data over time, and OxCGRT provides state-level policy information for defining mandate periods [7, 8]. The workflow includes data cleaning, harmonisation of survey and policy data, definition of pre-mandate and post-mandate periods, model training, model comparison, and interpretation of feature importance. Once the Australian workflow is established, the planned comparative analysis will extend to countries such as Brazil, Canada, China, India, the United Kingdom, and the United States, subject to final checks on data coverage and variable comparability. These countries were chosen because they differ in terms of geography, public health systems, pandemic ex-

perience, policy environment, and social context, while also maintaining overlap in the coverage of the YouGov Behaviour Tracker and OxCGRT data sources.

The project is guided by three research questions:

1. Within each country, do the key predictors of compliance with COVID-19 protective behaviours differ before and after sustained mask mandates?
2. Are these before–after differences similar across countries, or do they vary by national context?
3. Which predictors appear to be shared across countries, and which predictors are specific to particular national contexts?

2 Background

Previous research shows that compliance with COVID-19 protective behaviour is shaped by multiple individual and contextual factors rather than by policy rules alone. Demographic characteristics such as gender, age, and location have been linked to differences in mask wearing and other protective behaviours [2, 3]. Psychological and attitudinal factors, including perceived risk, efficacy beliefs, civic engagement, and broader attitudes toward protective measures, have also been associated with compliance [1]. In addition, cross-national research has shown that individual psychological variables and social norms can be important predictors of infection-prevention behaviour [4]. Together, these studies suggest that compliance reflects a combination of demographic, psychological, behavioural, and social factors.

A smaller body of work has used machine-learning methods to identify important predictors of COVID-19 infection-prevention behaviours and self-protective actions [1, 4, 5]. These approaches are useful because they can capture nonlinear relationships and interactions among predictors. However, much of the existing literature either focuses on single-country settings or examines policy effects without directly comparing whether predictors themselves change across policy periods and national contexts. Ryan et al. [6] provide direct evidence from Australia that mask mandates altered major determinants of adherence to protective health behaviours. The present project builds on this literature by using a consistent predictive framework to examine whether such before–after differences are also observable across countries.

3 Methods

This study builds on the methodological framework of Ryan et al. [6]. Several components of the data preparation, outcome construction, and model selection follow their approach, with adaptations made to suit the Australian state-level policy design used in the present analysis.

3.1 Data sources

This analysis combines individual-level Behaviour Tracker survey data with OxCGRT policy data. OxCGRT was used to derive state-specific sustained mask-mandate start dates from the H6 face-covering indicator. These dates were then assigned to survey responses by state and survey date, allowing each response to be classified as pre-mandate or post-mandate [7, 8].

3.1.1 YouGov COVID-19 Behaviour data

The first data source is the Imperial College London/YouGov COVID-19 Behaviour Tracker [7]. This dataset consists of repeated cross-sectional surveys conducted over time, meaning that different respondents were surveyed in different waves rather than the same individuals being followed longitudinally. The survey collected data on public behaviours and attitudes during the COVID-19 pandemic across multiple countries, including Australia, Brazil, Canada, China, India, the United Kingdom, and the United States.

The Behaviour Tracker contains anonymised respondent-level information on protective behaviours, demographic characteristics, perceived risk, trust in government, wellbeing, health-related variables, employment status, and related attitudes. Protective behaviour items include behaviours such as mask wearing, distancing, hand hygiene, isolation-related behaviour, and avoidance of risky settings. After cleaning, the Australian analytical dataset covers 2 September 2020 to 1 March 2022. Because the dataset is repeated cross-sectional, the analysis compares predictive patterns across responses and policy periods, but does not track behavioural change within the same individuals over time.

3.1.2 Oxford COVID-19 Government Response data

The second data source is the Oxford COVID-19 Government Response Tracker (OxCGRT) [8]. OxCGRT records government responses to COVID-19 over time and across jurisdictions. The dataset includes policy indica-

tors covering containment and closure policies, economic support, health-system measures, vaccination policies, and other government responses.

This project uses the OxCGRT mask-wearing indicator, identified in the dataset as policy variable H6, to define mask-mandate periods. The H6 indicator records face-covering policy on an ordinal scale, ranging from no policy or recommendation-only settings to requirements in specific public settings, requirements when social distancing is not possible, and requirements in all settings [8]. In this analysis, a value of 3 or above is treated as indicating a sustained mask-mandate context because it reflects a setting in which face coverings were required rather than merely recommended [6].

The Australian OxCGRT records provide daily policy information from March 2020 to January 2023. These records were used to identify the first sustained mask-mandate period for each Australian state. The resulting state-specific mandate dates were then used to classify survey responses as occurring before or after the relevant mandate period.

3.2 Data preparation pipeline

The Australian analysis was implemented as a structured data preparation pipeline designed to ensure consistency and reproducibility. The main steps include selecting predictors, defining outcomes, cleaning and recoding survey data, linking policy information, defining policy periods, and constructing training and test datasets. The full implementation is available in the project repository (Appendix A).

3.2.1 Predictor variables

Predictors were selected based on prior literature on COVID-19 behavioural compliance, the Australian study by Ryan et al. [6], and availability across survey waves and countries. The goal was to retain variables that are both theoretically relevant and comparable for future cross-national analysis.

The final predictors are grouped into six domains. Demographic variables include age, gender, employment status, household characteristics, and state. Trust-related variables capture confidence in government and evaluations of its response. Perceived illness-threat variables reflect perceived risk and severity of COVID-19. Health and wellbeing variables include mental-health indicators and self-reported health conditions. Behavioural variables capture willingness to isolate and engagement in non-mask protective behaviours. Temporal variables represent survey timing. Categorical variables were one-hot encoded with reference categories omitted.

3.2.2 Outcome variables

Two binary outcomes were constructed to represent protective behaviour compliance [6]. The first outcome captures face mask wearing, based on the median response to four mask-related items: mask use outside the home, in supermarkets, in retail settings, and on public transport. The second outcome represents broader protective behaviour, based on the median across available behaviour items including distancing, hygiene, and avoidance behaviours.

All behaviour items were measured on a five-point frequency scale. Median scores were converted into binary outcomes, where values of 4 or above indicate compliance and lower values indicate non-compliance [6].

3.2.3 Data cleaning and recoding

The Australian survey data were cleaned before linkage with the policy data. Missingness was first calculated for every variable in the raw Australian dataset, which contained 53 833 observations. Variables with a missing rate greater than 30% were removed (Appendix B). This threshold was selected to remove highly sparse variables while retaining theoretically important variables. Applying this rule removed 459 sparse variables and reduced the dataset from 513 variables to 54 variables.

Survey completion dates were converted into date format, and household size was recoded into numeric form. Responses of “8 or more” were coded as 8, while “Prefer not to say” and “Don’t know” were treated as missing values. During the period from 10 February 2021 to 18 October 2021, missing values in the PHQ4 mental-health variables and the d1 comorbidity variables were recoded as N/A following Ryan et al. [6].

After this adjustment, observations with remaining missing values were removed. This reduced the dataset from 53 833 observations to 36 349 observations. Ordinal perception variables were then converted into numeric scores. The variables r1_1 and r1_2 were recoded from a seven-point agreement scale into values from 1 to 7. Protective behaviour frequency responses were recoded onto a five-point scale, where “Not at all” was coded as 1, “Rarely” as 2, “Sometimes” as 3, “Frequently” as 4, and “Always” as 5 [6].

The face-mask behaviour score was calculated as the row-wise median of four mask-related behaviour items: mask use outside the home, in supermarkets, in retail settings, and on public transport. The broader protective behaviour score was calculated as the row-wise median of all available protective behaviour items. A separate non-mask protective behaviour score was also calculated by excluding the four mask-related items. Binary versions of the face-mask and broader protective behaviour

outcomes were created by coding scores of 4 or above as compliant and scores below 4 as non-compliant [6].

The original comorbidity indicators were collapsed into a single summary variable indicating whether the respondent reported comorbidities, no comorbidities, non-availability, or a preference not to answer. A fortnight-based time variable was also created from the survey completion date, with each value representing a 14-day period from the first survey date. After household-size conversion and removal of newly induced missing values, the cleaned dataset contained 35,460 observations and 26 variables.

The cleaned dataset was then converted into a modelling-ready format. The policy-period indicator was added to the cleaned survey data, and selected categorical variables were one-hot encoded using dummy variables with the first category omitted as the reference category. The encoded variables included state, gender, employment status, isolation-related responses, perceived government response, confidence in the health service, PHQ4 mental-health responses, and the collapsed comorbidity variable.

The preprocessing steps, after encoding, expanded the dataset from 26 variables to 65 variables and 35 460 observations. The Variable names were then changed for more readable.

3.2.4 Policy period definition

Survey observations were classified as occurring before or after the first sustained state-level mask-mandate onset using the OxCGRT mask-wearing indicator. Because mask policies varied across states, the classification was defined separately for each state.

A 14-day rolling average of the mask-wearing indicator was used to identify sustained policy changes [6]. A state was classified as entering a sustained mask-mandate period when this rolling average first reached 3 or above. Each observation was then assigned to the pre-mandate or post-mandate-onset period based on its survey date and state.

3.2.5 Data split and class balance

Four modelling tasks were created by combining the two binary outcomes with the two policy periods: face-mask wearing and broader protective behaviour, each assessed before and after the mandate. Each task was split independently into training and test sets using an 80/20 split, with stratification used where possible to preserve outcome class balance.

Predictor sets were defined separately for the face-mask and protective behaviour tasks. Identifiers, dates, policy-period indicators, and

target-related variables were excluded. For the protective behaviour tasks, the non-mask protective behaviour score was also excluded to avoid target leakage [6].

3.3 Models

3.3.1 Logistic regression

Logistic regression was used as a baseline linear model [6, 9]. It estimates the probability of compliance as a function of predictors on the logit scale, providing a simple and interpretable benchmark.

3.3.2 Classification tree

A classification tree was used to model nonlinear relationships through recursive partitioning of the predictor space [6, 10]. The model produces decision rules that segment observations into homogeneous outcome groups.

3.3.3 Random forest

Random forest extends decision trees by combining multiple trees built on bootstrapped samples with random feature selection [6, 11]. This approach reduces variance and improves predictive stability while capturing nonlinear relationships and interactions.

3.3.4 XGBoost

XGBoost is a gradient boosting method that builds trees sequentially to minimise prediction error [6, 12]. Each new tree is fitted using information from previous trees, allowing the ensemble to capture complex patterns in the data. This model is included for its strong predictive performance and its ability to provide feature importance measures.

3.4 Evaluation

Model performance was evaluated using accuracy, precision, recall, F1 score, and ROC AUC. ROC AUC was used as the primary metric because it is threshold-independent and useful when class balance differs across tasks. Five-fold cross-validation results were summarised as mean performance with standard error, calculated as the standard deviation across folds divided by $\sqrt{5}$. Because hyperparameter tuning was performed on the training data, cross-validation results were used mainly

for model selection. Final performance was then reported on the held-out test set for each modelling task.

4 Results

4.1 Cross-validation model comparison

Table 1 reports 5-fold cross-validation performance for the four candidate models across the four outcome-period tasks.

Table 1. 5-fold cross-validation results for logistic regression, classification tree, random forest, and XGBoost across the four Australian modelling tasks. Values are reported as mean (standard error), where standard errors were calculated across the five validation folds.

Model	AUC	Precision	Recall	Accuracy	F1
Before mandate – Face mask					
Logistic regression	0.821 (0.006)	0.510 (0.007)	0.774 (0.010)	0.729 (0.006)	0.615 (0.007)
Classification tree	0.819 (0.007)	0.516 (0.005)	0.736 (0.008)	0.734 (0.004)	0.607 (0.006)
Random forest	0.857 (0.005)	0.616 (0.009)	0.701 (0.014)	0.794 (0.005)	0.655 (0.009)
XGBoost	0.864 (0.004)	0.639 (0.013)	0.676 (0.010)	0.802 (0.006)	0.656 (0.008)
After mandate – Face mask					
Logistic regression	0.836 (0.004)	0.879 (0.004)	0.780 (0.004)	0.769 (0.003)	0.826 (0.003)
Classification tree	0.859 (0.003)	0.883 (0.001)	0.863 (0.006)	0.823 (0.005)	0.873 (0.004)
Random forest	0.890 (0.003)	0.890 (0.003)	0.893 (0.003)	0.847 (0.004)	0.892 (0.003)
XGBoost	0.893 (0.004)	0.892 (0.003)	0.905 (0.003)	0.856 (0.004)	0.899 (0.003)
Before mandate – Protective behaviour					
Logistic regression	0.733 (0.002)	0.658 (0.004)	0.699 (0.006)	0.673 (0.004)	0.678 (0.004)
Classification tree	0.740 (0.006)	0.667 (0.007)	0.706 (0.018)	0.682 (0.008)	0.686 (0.010)
Random forest	0.785 (0.003)	0.704 (0.005)	0.711 (0.009)	0.710 (0.003)	0.707 (0.003)
XGBoost	0.788 (0.004)	0.706 (0.005)	0.703 (0.004)	0.709 (0.004)	0.705 (0.003)
After mandate – Protective behaviour					
Logistic regression	0.778 (0.002)	0.842 (0.001)	0.727 (0.002)	0.715 (0.002)	0.780 (0.001)
Classification tree	0.791 (0.002)	0.861 (0.007)	0.692 (0.022)	0.707 (0.009)	0.766 (0.011)
Random forest	0.835 (0.002)	0.854 (0.002)	0.814 (0.002)	0.773 (0.001)	0.833 (0.001)
XGBoost	0.844 (0.002)	0.861 (0.002)	0.820 (0.002)	0.782 (0.002)	0.840 (0.001)

The highest cross-validated ROC AUC was observed for post-mandate face mask wearing, where XGBoost achieved an AUC of 0.893. The lowest XGBoost ROC AUC was observed for pre-mandate broader protective behaviour, where the corresponding value was 0.788. XGBoost achieved the highest mean ROC AUC in all four tasks. Based on cross-validated ROC AUC, XGBoost was selected as the final model for all four tasks.

4.2 Held-out test performance

Table 2 reports test performance for the selected XGBoost model.

Table 2. Held-out test performance for the selected XGBoost models across the four Australian modelling tasks. XGBoost was selected for all four tasks based on cross-validated ROC AUC.

Selected model	AUC	Precision	Recall	Accuracy	F1
Before mandate – Face mask					
XGBoost	0.866	0.622	0.675	0.794	0.647
After mandate – Face mask					
XGBoost	0.898	0.89	0.895	0.848	0.892
Before mandate – Protective behaviour					
XGBoost	0.783	0.69	0.703	0.698	0.696
After mandate – Protective behaviour					
XGBoost	0.848	0.856	0.811	0.773	0.833

The held-out test results show a similar pattern to the cross-validation results. Test-set ROC AUC was higher after mandates than before mandates for both outcomes: 0.898 compared with 0.866 for face mask wearing, and 0.848 compared with 0.783 for broader protective behaviour. The face-mask models also achieved higher AUC values than the corresponding broader protective behaviour models in both policy periods. For the pre-mandate face-mask task, precision was 0.622 and recall was 0.675, compared with precision of 0.89 and recall of 0.895 after mandates.

The ROC curves for the four models are provided in Appendix C, which visually supplement the test-set ROC AUC values reported in Table 2.

4.3 Feature importance

Figure 1 reports XGBoost feature importance with state indicators included. In the face-mask model, state indicators are prominent in both periods. New South Wales has the largest before-mandate feature importance value among the plotted predictors, while Western Australia has the largest after-mandate value. For broader protective behaviour, Western Australia has the largest after-mandate feature importance value among the plotted predictors, while New South Wales has the largest before-mandate value.

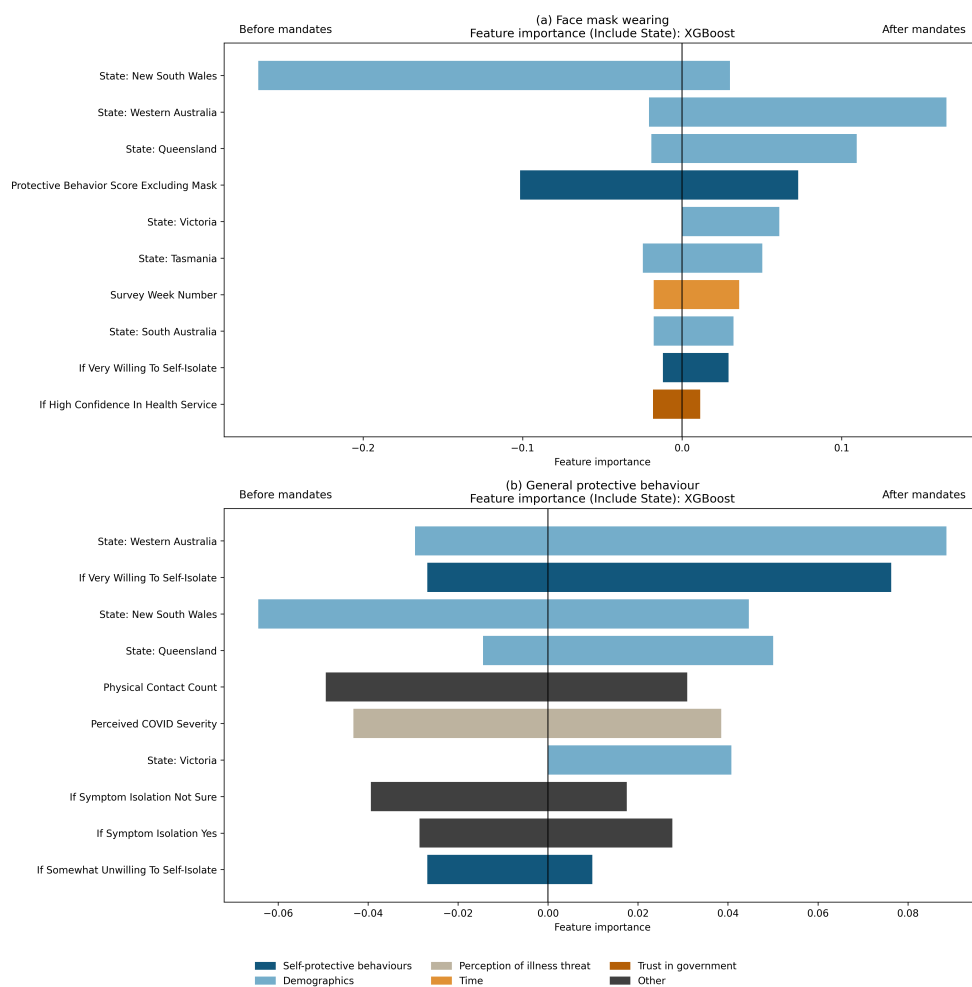


Figure 1. XGBoost feature importance (including State). Panel (a) shows face mask predictors, and panel (b) shows broader protective behaviour. Bars represent feature importance before and after mask mandates, with predictors on the y-axis and relative importance on the x-axis. Colours indicate predictor domains

Table 3. Top 10 XGBoost feature importance values including state indicators. Percentages represent feature importance values for the pre-mandate and post-mandate models.

Outcome	Feature	Category	Before/After (%)
Face mask wearing			
	If High Confidence In Health Service	Trust in government	1.83 / 1.14
	If Very Willing To Self-Isolate	Self-protective behaviours	1.21 / 2.90
	State: South Australia	Demographics	1.77 / 3.23
	Survey Week Number	Time	1.77 / 3.58
	State: Tasmania	Demographics	2.46 / 5.02
	State: Victoria	Demographics	0.00 / 6.10
	Protective behaviour Score Excluding Mask	Self-protective behaviours	10.16 / 7.28
	State: Queensland	Demographics	1.93 / 10.95
	State: Western Australia	Demographics	2.08 / 16.58
	State: New South Wales	Demographics	26.56 / 3.00
broader protective behaviour			
	If Somewhat Unwilling To Self-Isolate	Self-protective behaviours	2.68 / 0.98
	If Symptom Isolation Yes	Other	2.86 / 2.77
	If Symptom Isolation Not Sure	Other	3.94 / 1.75
	State: Victoria	Demographics	0.00 / 4.07
	Perceived COVID Severity	Perception of illness threat	4.33 / 3.85
	Physical Contact Count	Other	4.94 / 3.09
	State: Queensland	Demographics	1.45 / 5.01
	State: New South Wales	Demographics	6.44 / 4.46
	If Very Willing To Self-Isolate	Self-protective behaviours	2.69 / 7.63
	State: Western Australia	Demographics	2.96 / 8.86

Table 3 shows that, when state indicators are included, state-related variables account for several of the largest feature importance values. For face mask wearing, State: New South Wales has the highest pre-mandate value at 26.56%, while State: Western Australia has the highest post-mandate value at 16.58%. Other state indicators also appear among the top 10 predictors, including Queensland, Victoria, Tasmania, and South Australia. Protective behaviour Score Excluding Mask is the largest non-state predictor for face mask wearing, with values of 10.16% before mandates and 7.28% after mandates.

For broader protective behaviour, State: Western Australia has the largest post-mandate feature importance value at 8.86%, while State: New South Wales has the largest state-related pre-mandate value at 6.44%. Physical Contact Count has the largest non-state pre-mandate value at 4.94%, followed by Perceived COVID Severity at 4.33% and If Symptom Isolation Not Sure at 3.94%. After mandates, If Very Willing To Self-Isolate has a feature importance value of 7.63%, which is the largest non-state value in the post-mandate broader protective behaviour model.

Figure 2 shows XGBoost feature importance after excluding the “state” variable. For face mask wearing, the protective behaviour score excluding mask was the strongest non-state predictor in both periods, although its importance was higher before mandates. Survey week number and willingness to self-isolate became more important after mandates. Age, comorbidity, trust in health services, and employment status contributed more before mandates. For broader protective behaviour, physical contact count was the strongest pre-mandate predictor, while willingness to self-isolate was strongest after mandates.

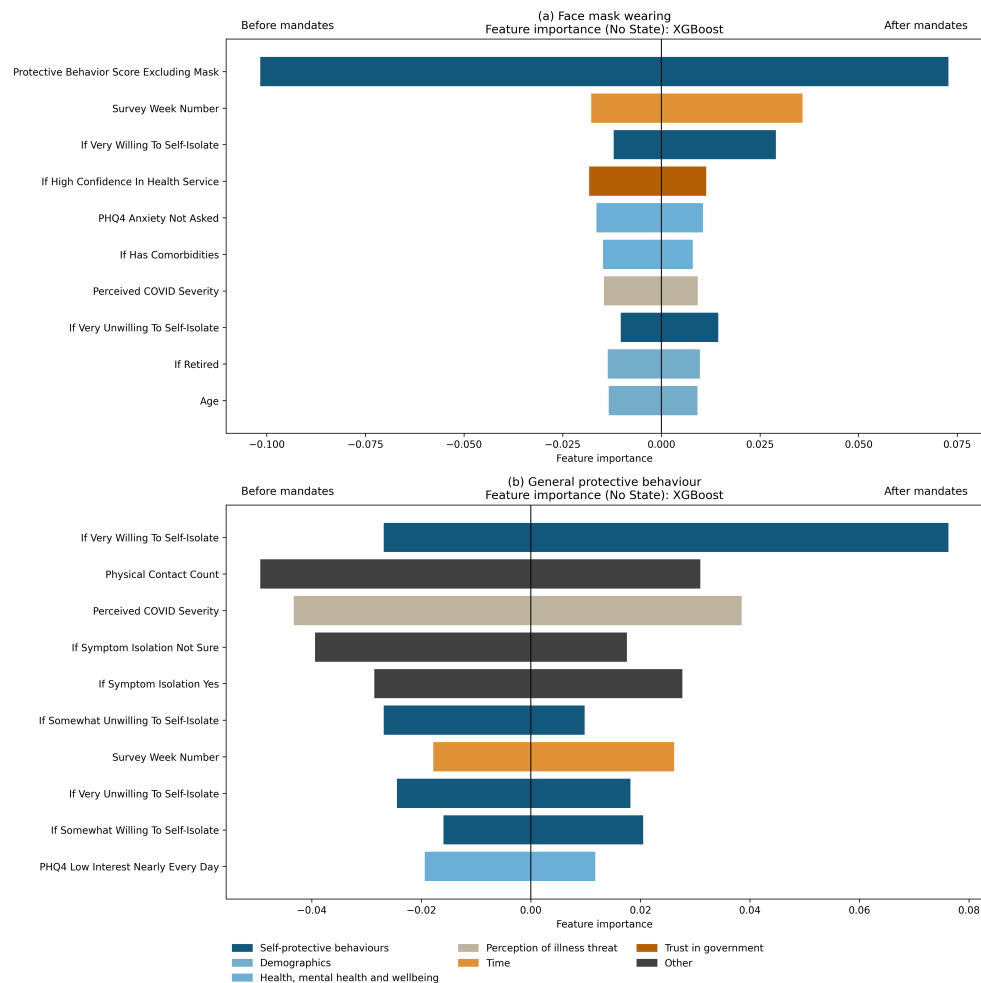


Figure 2. XGBoost feature importance after excluding state indicators. Panel (a) compares non-state predictors of face mask wearing before and after sustained mask mandates. Panel (b) shows the corresponding comparison for broader protective behaviour. Left-hand bars show the pre-mandate model and right-hand bars show the post-mandate model.

Table 4. Top 10 XGBoost feature importance values after excluding state indicators. Percentages represent feature importance values for the pre-mandate and post-mandate models.

Outcome	Feature	Category	Before/After (%)
Face mask wearing			
	Age	Demographics	1.33 / 0.92
	If Retired	Demographics	1.36 / 0.98
	If Very Unwilling To Self-Isolate	Self-protective behaviours	1.03 / 1.44
	Perceived COVID Severity	Perception of illness threat	1.45 / 0.93
	If Has Comorbidities	Health, mental health and wellbeing	1.48 / 0.80
	PHQ4 Anxiety Not Asked	Health, mental health and wellbeing	1.65 / 1.05
	If High Confidence In Health Service	Trust in government	1.83 / 1.14
	If Very Willing To Self-Isolate	Self-protective behaviours	1.21 / 2.90
	Survey Week Number	Time	1.77 / 3.58
	Protective behaviour Score Excluding Mask	Self-protective behaviours	10.16 / 7.28
broader protective behaviour			
	PHQ4 Low Interest Nearly Every Day	Health, mental health and wellbeing	1.94 / 1.17
	If Somewhat Willing To Self-Isolate	Self-protective behaviours	1.60 / 2.05
	If Very Unwilling To Self-Isolate	Self-protective behaviours	2.45 / 1.82
	Survey Week Number	Time	1.78 / 2.62
	If Somewhat Unwilling To Self-Isolate	Self-protective behaviours	2.68 / 0.98
	If Symptom Isolation Yes	Other	2.86 / 2.77
	If Symptom Isolation Not Sure	Other	3.94 / 1.75
	Perceived COVID Severity	Perception of illness threat	4.33 / 3.85
	Physical Contact Count	Other	4.94 / 3.09
	If Very Willing To Self-Isolate	Self-protective behaviours	2.69 / 7.63

Table 4 shows the top feature importance values after excluding state indicators. For face mask wearing, Protective behaviour Score Excluding Mask remains the largest predictor in both periods, with values of 10.16% before mandates and 7.28% after mandates. Survey Week Number increases from 1.77% before mandates to 3.58% after mandates, and If Very Willing To Self-Isolate increases from 1.21% to 2.90%. Several health, mental-health, demographic, and trust-related predictors have higher pre-mandate than post-mandate values.

For broader protective behaviour, If Very Willing To Self-Isolate has the highest post-mandate value at 7.63%, while Physical Contact Count has the highest pre-mandate value at 4.94%. Perceived COVID Severity has values of 4.33% before mandates and 3.85% after mandates. If Symptom Isolation Not Sure decreases from 3.94% before mandates to 1.75% after mandates, while Survey Week Number increases from 1.78% to 2.62%.

5 Discussion

This report examines whether the predictors of COVID-19 protective behaviour changed before and after Australia implemented sustained mask mandates. The Australian results suggest that the predictive structure of compliance differed between the pre-mandate and post-mandate periods. After mandates, both face mask wearing and broader protective behaviour were predicted more accurately, and the main predictors also changed across the two periods. This indicates that the factors associated with protective behaviour were not static, but varied across the pre-mandate and post-mandate policy periods.

For face mask wearing, model performance improved after mandates. The XGBoost test ROC AUC increased from 0.866 before mandates to 0.898 after mandates, while precision increased from 0.622 to 0.890 and recall increased from 0.675 to 0.895. This indicates that the predictors captured mask-wearing behaviour more effectively in the post-mandate period. The feature-importance results are consistent with this interpretation. When state indicators were included, pre-mandate face mask wearing was dominated by State: New South Wales, with a feature importance value of 26.56%. After mandates, this fell to 3.00%. In contrast, State: Western Australia increased from 2.08% to 16.58%, State: Queensland increased from 1.93% to 10.95%, and State: Victoria increased from 0.00% to 6.10%. Therefore, state indicators made an important predictive contribution both before and after the introduction of sustained mask mandates, although the most informative states changed across the two periods. These state indicators are not simply geographical factors, but are more likely a collection of various influencing factors in different regions, including the intensity of enforcement of mandates, local transmission, law enforcement environment, and other unmeasured regional differences [6]. The non-state predictors of face mask wearing also changed. Before mandates, the strongest non-state predictor was Protective Behaviour Score Excluding Mask, with a feature importance value of 10.16%. After mandates, it remained important but decreased to 7.28%. At the same time, Survey Week Number increased from 1.77% to 3.58%, and If Very Willing To Self-Isolate increased from 1.21% to 2.90%. This indicates that within the pre-mandate model, mask wearing was most strongly associated with existing engagement in other protective behaviours. After mandates, mask wearing became more connected with pandemic timing and willingness to follow isolation-related public health advice.

For broader protective behaviour, the XGBoost test ROC AUC increased from 0.783 before mandates to 0.848 after mandates, and F1

increased from 0.696 to 0.833. Before mandates, the most important non-state predictors were Physical Contact Count at 4.94%, Perceived COVID Severity at 4.33%, and If Symptom Isolation Not Sure at 3.94%. These results show that broader protective behaviour was initially predicted more by contact patterns, perceived illness threat, and uncertainty about isolation. After mandates, If Very Willing To Self-Isolate became the strongest non-state predictor, increasing from 2.69% to 7.63%. This is the clearest individual-level change in the analysis: willingness to self-isolate changed from a moderately important predictor before mandates to the dominant non-state predictor after mandates. This finding means that after mask mandates, broader protective behaviour was more closely related to a general willingness to follow restrictive public health advice. In the post-mandate model, perceived severity and physical contact patterns were no longer the strongest non-state predictors. Instead, broader protective behaviour was most strongly associated with whether respondents were willing to self-isolate. This supports previous research showing that compliance is shaped not only by demographics and risk perception, but also by attitudes, behavioural intentions, and social context [1–5].

Taken together, the Australian results answer the first research question clearly. The key predictors of compliance differed before and after sustained mask mandates. For face mask wearing, the main change was from a pre-mandate pattern strongly influenced by State and existing non-mask protective behaviour, to a post-mandate pattern in which State, survey timing, and willingness to self-isolate became more important. For broader protective behaviour, the main change was from physical contact, perceived COVID severity, and symptom-isolation uncertainty before mandates, to willingness to self-isolate after mandates.

The main strength of this work is that it converts the Australian analysis into a reproducible modelling pipeline. The workflow defines comparable outcomes, constructs state-specific mandate periods, applies stratified train–test splitting, compares multiple classifiers, evaluates the selected model on held-out test data, and examines feature importance with and without state indicators. This provides a practical foundation for applying the same framework to other countries.

There are also limitations. First, only Australia has been analysed so far, so the present results cannot show whether the same before–after changes appear internationally. Second, the Behaviour Tracker is repeated cross-sectional rather than longitudinal, so the analysis compares groups of responses over time rather than tracking the same individuals. Third, the pre-mandate and post-mandate classification simplifies complex policy histories across states. Fourth, feature importance measures predictive contribution, not causal influence. Therefore, the results

show that the predictive structure changed after mandates, but they do not prove that mandates caused each predictor to become more or less important.

Overall, the Australian case demonstrates that the predictors of mask-wearing behaviour and broader protective behaviour differed before and after the introduction of sustained mask mandates in Australia. The next stage is to apply the same pipeline to Brazil, Canada, China, India, the United Kingdom, and the United States to assess whether similar before–after changes appear across national settings.

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A All code and data

The analysis code is available in the project GitHub repository:

<https://github.com/ZixaunZhao/Covid-mask-compliance-analysis.git>

The main notebook is `ALL.ipynb`, which contains the integrated workflow for data preparation, model fitting, evaluation, and result generation. The repository also contains separate notebooks for data cleaning, data splitting, logistic regression, classification tree, random forest, and XGBoost analyses.

B Missingness threshold selection

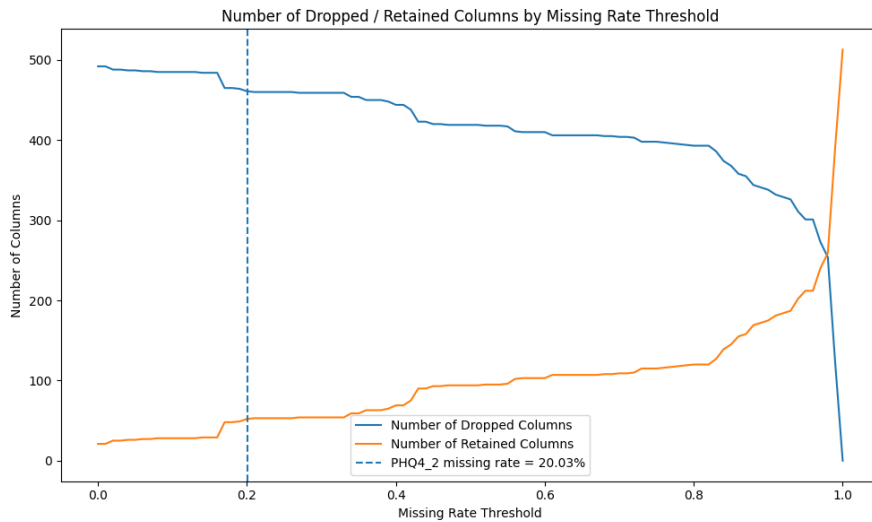


Figure 3. Number of dropped and retained variables under alternative missing-rate thresholds. The dashed vertical line indicates the missing rate of PHQ4_2 at 20.03%. A 30% missing-rate threshold was selected because it is above this value while still removing highly sparse variables.

C ROC curves

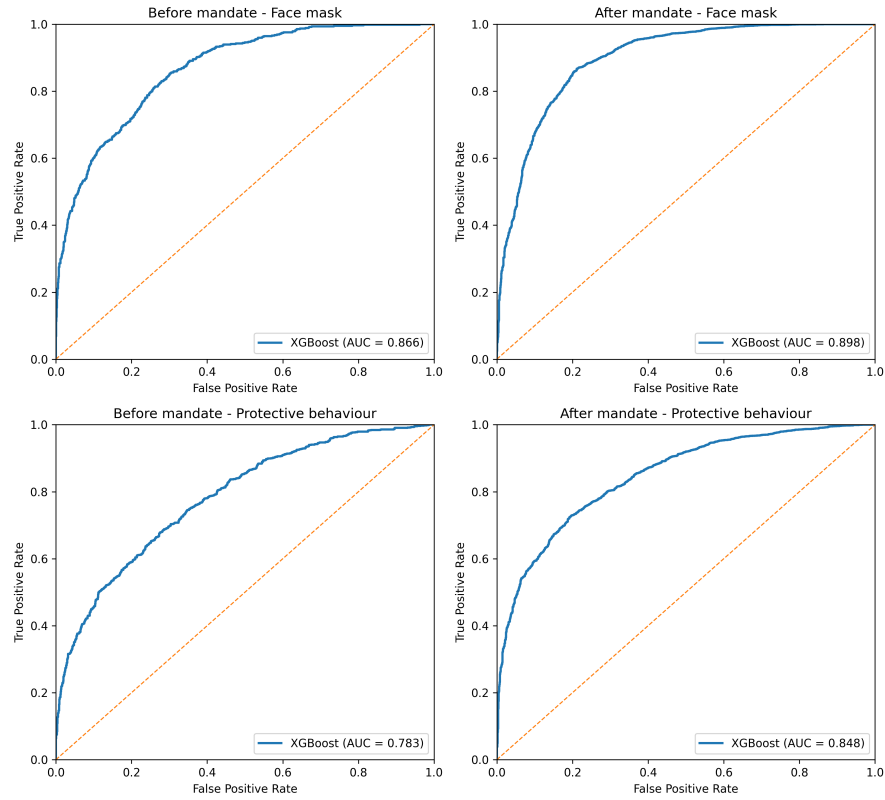


Figure 4. ROC curves for the selected XGBoost models across the four Australian modelling tasks. The curves provide a visual supplement to the held-out test ROC AUC values reported in Table 2.

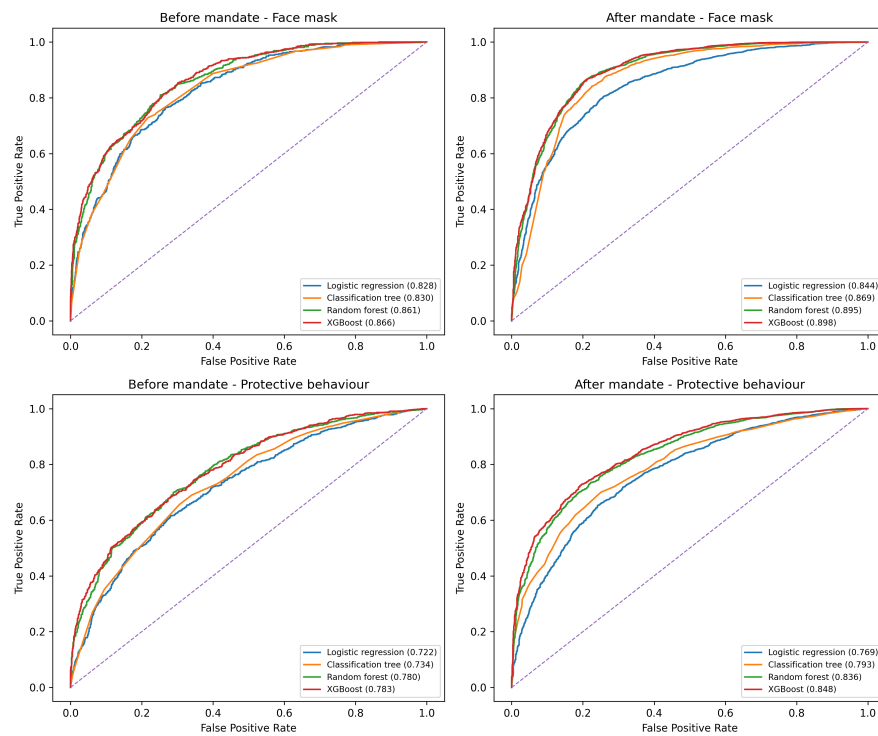


Figure 5. ROC curves for all four modelling algorithms (Logistic regression, Classification tree, Random forest, XGBoost) across the four Australian modelling tasks